# TideTail: Evaluating Extreme-Aware Diffusion Augmentation for Coastal Water-Level Forecasting

Yingda Yu Tong Mengxue Rong Jia

October 30, 2026
*Preprint — evidence-linked, not all hypotheses verified*

**Abstract**

Coastal water-level forecasting faces a fundamental data scarcity problem: the most operationally critical extreme storm events are rare in historical records, yet predicting them correctly is the primary goal of coastal risk management. We investigate whether conditional diffusion models can generate physically plausible extreme coastal water-level residual samples that improve downstream forecasting on independent real storm events, under matched forecaster training budgets. This paper reports the pre-registration of this investigation, the frozen data protocol, the engineering implementation, and the negative result of our first controlled diffusion pilot. The 27,169-parameter conditional DDPM trained for three epochs on real residual data produces samples whose standard deviation is approximately **22.95 times** that of the training target, with magnitudes exceeding $\pm 16$ metres. We trace this failure through independent CPU reproduction, terminal-step noise sensitivity analysis ($\alpha_{\text{bar}} \approx 5.93 \times 10^{-7}$), and end-to-end audit of the forward and reverse stochastic equations. The diagnosis confirms that while the model architecture and diffusion mathematics are formally correct, terminal-step amplification combined with imperfect epsilon prediction at the training budget causes catastrophic scale explosion. We present this finding, our repaired data-splitting infrastructure, and our frozen research protocol as a methodological contribution to the study of diffusion models for extreme-event time-series generation.

## 1 Introduction

Coastal flooding causes approximately $1 trillion in global damages annually and displaces hundreds of thousands of people [Yang et al., 2023]. Reliable forecasting of extreme water levels during storm events is therefore an urgent public safety problem. The core challenge is *data imbalance*: the most operationally critical storm surges are also the rarest. Training standard neural forecasters on historically recorded water levels allocates most model capacity to predicting the frequent benign conditions rather than the infrequent but catastrophic extremes.

Data augmentation has emerged as a promising approach to rebalance training distributions. For time-series forecasting, augmentation traditionally takes the form of oversampling, bootstrapping, noise perturbation, or interpolation. These methods are well-established for image data but have received less scrutiny for coastal water-level prediction, where physical laws (tidal cycles, wave dynamics, storm surge propagation) impose strong constraints on what constitutes a plausible augmented sample.

Conditional diffusion models [Ho et al., 2020] offer a generative framework that could produce physically realistic extreme events by learning the conditional distribution of water-level residuals given historical gauge observations. The hypothesis driving this work is:

> **H1**: Tail-aware conditional diffusion augmentation, applied under a matched forecaster training budget, improves prediction accuracy on *independent held-out real storm events* relative to real-only training, standard oversampling, perturbation augmentation, and standard conditional diffusion.

A secondary hypothesis (H2) investigates whether adding physical consistency or multi-gauge spatial coherence further improves synthesis usefulness. Both are *unverified hypotheses* at this writing. This paper reports the frozen research protocol, the engineering implementation, and the first controlled empirical evidence — which is, at this stage, a *negative result*: our pilot DDPM produces samples of physically implausible magnitude. We argue that this finding, its diagnosis, and the methodological infrastructure built to obtain it are themselves a contribution to the study of diffusion models for extreme-event time-series generation.

### 1.1 Contributions

1. We present a fully documented, leakage-safe research protocol for coastal water-level forecasting with diffusion augmentation, including event-grouped chronological splits, full-support embargo, train-only thresholds, and a six-region data manifest frozen at license-admission boundaries.
2. We implement and audit a conditional denoising diffusion probabilistic model (DDPM) for residual water-level generation, with 27,169 parameters, a cosine noise schedule, and 64 reverse diffusion steps.
3. We report a *reproduced negative result*: independently verified on both CUDA and CPU, the model's 24-hour generated residuals span $-16.91$ to $+7.94$ metres (SD 4.47m), approximately $22.95\times$ the training target standard deviation of 0.215m. We diagnose terminal-step amplification and imperfect epsilon prediction as contributing factors.
4. We present engineering validation guards (nonfinite rejection, train-reference scale screen) that prevent absurdly bad samples from being misclassified as successful generation. These guards are validation infrastructure, not model repairs.

### 1.2 Paper structure

Section 2 surveys related work. Section 3 details our frozen research protocol and data admission. Section 4 describes the conditional DDPM architecture and training setup. Section 5 describes our experimental plan and implementation. Section 6 reports our empirical findings, including the negative result of the pilot. Section 7 discusses implications, limitations, and ongoing work. Section 8 concludes.

## 2 Related Work

### 2.1 Coastal water-level forecasting

Traditional coastal water-level forecasting combines deterministic tidal constituents with empirical or dynamical storm-surge models [Yang et al., 2023]. In recent years, neural approaches have

improved forecast quality on held-out gauges. Sergeev et al. [2024] demonstrated that information from multiple gauges along a coastline substantially improves forecast accuracy for downstream stations. Tong et al. [2024] introduced dual-timescale residual memory that separately models slow tidal evolution and fast storm-driven residuals, with particular attention to extreme events. These works establish that (a) the de-tided residual signal carries the most operationally relevant information during storms, and (b) extreme events demand model architectures that differ from those optimized for benign conditions.

### 2.2 Data augmentation for time-series forecasting

Data augmentation for time series is reviewed by Hoppe et al. [2024] and related surveys. Standard approaches include: (1) *oversampling* rare events or weighting them in the loss function, (2) *perturbation* with additive or multiplicative noise, (3) *interpolation* between adjacent real windows, and (4) *generative models* including GANs, VAEs, and diffusion. A key distinction for coastal water levels is whether augmentation preserves the *physical plausibility* of the residual signal: unrealistic samples may actually degrade downstream forecasting by corrupting the training distribution.

### 2.3 Conditional diffusion models for time series

Conditional score-based diffusion models have been applied to time-series imputation [Tashiro et al., 2021, Ma et al., 2023] and generation. The canonical formulation [Ho et al., 2020] defines a forward process $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$ and trains a network $\epsilon_\theta(x_t, t, c)$ to predict the injected noise, where $c$ denotes conditioning information. During generation, the reverse process iteratively denoises from $T$ back to 0:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t, c)\right) + \sigma_t z, \tag{1}$$

where $\sigma_t^2 = \beta_t(1-\bar{\alpha}_{t-1})/(1-\bar{\alpha}_t)$ is the posterior variance and $z \sim \mathcal{N}(0, I)$.

### 2.4 This work's niche

To our knowledge, this is the first systematic application of conditional diffusion augmentation to coastal water-level forecasting with (a) event-grouped leakage-safe splits, (b) train-only threshold estimation, (c) matched-budget comparison across augmentation methods, and (d) pre-registration of hypotheses before running experiments.

## 3 Research Protocol

### 3.1 Data source and admission

The authorized owner copy of the dataset is at `/data/yingda/tidetail-data/prime-formal-2015-2024`, containing 350 files totaling 451 MB. Six coastal regions are included, with 87,672 hourly UTC rows each spanning 2015-01-01 to 2024-12-31. Region admission status is recorded in `region_availability.json`:

The Hong Kong region is excluded from all formal experiments until the data owner confirms license, vertical datum, and the HK event `pearl_river_hong_kong_event_01149`, which spans the validation/test boundary at index 78904. The migration manifest (`MIGRATION_MANIFEST.json`) exists one level above the release directory; file SHA256 and size checks confirm all inspected files match both release and migration inventory entries.

Table 1: Region formal training authorization status

| Region | Training authorized |
|---|---|
| New York Bight | Yes |
| Thames / North Sea | Yes |
| Hong Kong Pearl River | **No** (license/datum not signed) |
| Southeast Australia | Yes |
| US Gulf Coast | Yes |
| Charleston Atlantic | Yes |

### 3.2 Target definition

The release manifest `database_manifest.json` defines:

- **Total water level**: observed water level above local chart datum.
- **Tide**: deterministic FES2022 tidal constituent forecast (no future uncertainty in this archival release).
- **Residual**: water_level_m = observed_total − FES2022_tide.

The pilot diffusion model targets `water_level_m` (residual), in metres, with a fixed 72-hour history → 24-hour forecast window for engineering feasibility. The formal release contract specifies 96-hour history → 72-hour forecast; all comparisons between methods should use one consistent contract for scientific validity.

### 3.3 Leakage-safe splitting

Our protocol enforces three hard gates before any window creation or normalization:

1. **Event-group disjointness**: the same independent storm event may not appear in two partitions. Events are represented as `EventGroup(event_id, start_idx, end_idx)` intervals. The implementation in `src/tidetail/splits.py` validates non-overlapping intervals and unique IDs before splitting.
2. **Chronological ordering**: train row block must end before validation row block, which must end before test row block. The split is 8:1:1 by chronological event-group order, subject to `min_events_per_split`.
3. **Full-support embargo**: for forecast origin time $o$, lookback $L$, and horizon $H$, the full temporal support of one forecast window is $[o-L, o+H)$. Two partitions must not have overlapping full-support intervals. The repaired `embargo_filter` compares every index in an event group to every protected index, replacing the earlier midpoint-only comparison that incorrectly accepted dangerous events (see Appendix A). Additionally, `embargo_full_support_filter` enforces $o \geq$ split_block_start $+ L$ and $o \leq$ split_block_end $- H$ for each forecast origin.

All scaler fitting, quantile threshold estimation, and interpolation fitting are performed on *train rows only* and then applied to all partitions without re-fitting. Threshold-run events crossing the train boundary are removed in full before window creation, not partial-interval trimmed.

### 3.4 Formal release contract

The database manifest specifies:

- **Input**: 96 hours of historical observations.
- **Horizon**: 72 hours of forecast.
- **Split**: chronological 8:1:1 with strict window containment.
- **Tide**: existing local FES2022 deterministic archive.
- **Forcing**: ERA5 hindcast only. No operational 72-hour forecast claim without archived issue/valid/lead NWP forecast vintages.

The ERA5 forcing data available in this release is historical hindcast data, not forecast data. Using future observed ERA5 values as conditioning information in a forecast experiment would constitute leakage unless explicitly labelled as an oracle experiment. Our gauge-only diffusion model does not consume atmospheric forcing; this limitation is discussed in Section 7.

## 4 Method

### 4.1 Conditional DDPM architecture

We implement a 1D temporal denoising diffusion probabilistic model with the following components:

1. **Temporal embedding**: sinusoidal frequencies $f_j = 10000^{-2j/D}$ embedded into a width-`width` MLP, where $D$ is half the width.
2. **History conditioning**: a separate MLP consumes all 72 past residuals, producing a `width`-dimensional conditioning vector that is *added* to the temporal embedding before entering the denoiser.
3. **Dilated residual blocks**: four blocks with dilation $\{1, 2, 4, 8\}$, each containing GroupNorm (groups=4), SiLU activation, 3-conv padding to preserve length, and a linear condition injection.
4. **Output head**: GroupNorm + SiLU + 3-conv producing a single-channel epsilon prediction at each of the 24 forecast horizons.

Total parameter count: 27,169.

### 4.2 Forward and reverse processes

We implement cosine cumulative noise schedule [Ho et al., 2020] with 64 diffusion steps:

$$\beta_t = 1 - \frac{\cos((t + 0.008)/1.008 \cdot \pi/2)^2}{\cos((t - 1 + 0.008)/1.008 \cdot \pi/2)^2}, \qquad \alpha_t = 1 - \beta_t, \qquad \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i. \tag{2}$$

### 4.3 Training

Training targets *unit Gaussian epsilon* at each timestep. The loss is mean squared error between the network's epsilon prediction and the actual injected Gaussian noise:

$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon\sim\mathcal{N}(0,I)}\left[\left\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t, \text{history})\right\|^2\right]. \tag{3}$$

Training hyperparameters for the pilot run (Issue #6): AdamW with lr = 0.001, batch size 128, three epochs, BF16 precision on RTX3090 GPU 4, seed 20261025.

### 4.4 Engineering validation guards

After the pilot revealed catastrophic scale explosion, we added validation guards that *reject* nonfinite or implausibly scaled samples. These are engineering checks, not model repairs:

1. **Nonfinite rejection**: raise on nonfinite conditioning history, nonfinite epsilon prediction at any reverse step, or nonfinite reverse sample intermediate state.
2. **Train-reference scale screen**: after de-normalization, compare sample standard deviation to frozen training target standard deviation. Reject if ratio $< 0.1$ (collapse) or $> 5.0$ (dispersion), or if mean offset exceeds 5 training standard deviations. Both bounds are explicit policy defaults, not estimates of physical plausibility.

The historical GPU pilot predates these guards; its samples have SD ratio 22.95 and would be rejected by the screen.

## 5 Experiments

### 5.1 Comparison groups

Following the pre-registered protocol, the five comparison groups are:

- **A0**: Real-only training, no augmentation.
- **A1**: Oversampled or balanced-loss training on real data (not prediction-noise augmentation).
- **A2**: Physically justified perturbation or interpolation.
- **A3**: Standard conditional diffusion (no tail-awareness).
- **A4**: Tail-aware conditional diffusion (the primary H1 hypothesis).

All groups share the same downstream predictor architecture, total optimizer steps, training sample budget, hyperparameter tuning budget, and random seeds. All test evaluations use *only independent held-out real storm events*. A proposed A5 (A4 + physical/spatial constraints) is conditional on positive A4 evidence.

### 5.2 Pilot experiment (Issue #6)

The pilot investigates the *generative model itself*, without downstream forecaster evaluation. It uses a frozen 72-hour $\rightarrow$ 24-hour window contract on New York Bight gauge NOAA_8518750_The_Battery, training on 69,946 overlapping windows from 70,041 finite training residuals. The normalization mean is 0.108 m and SD is 0.215 m, fitted before window creation on train rows only.

### 5.3 Evaluation metrics

The pilot records engineering diagnostics rather than forecast utility:

- Per-step epsilon MSE, first and final step loss, first/final 100-step means.
- Generated sample magnitude (min, max, SD) after inverse normalization.
- Sample scale screen outcome (finite, std ratio, mean offset).
- Terminal-step alpha_bar and epsilon-to-clean amplification factor.
- Condition sensitivity (standardized 72-hour history shift) and nearest-training-window RMSE (eight probes, not a full memorization audit).

### 5.4 Hardware and run environment

The GPU pilot executed on RTX3090 GPU index 4 (UUID `GPU-6be030f8-625e-c4e7-a2fb-1309a673664b`) with `CUDA_VISIBLE_DEVICES=4` and explicit BLAS/OpenMP thread caps at 2. The checkpoint (444,971 bytes, SHA256 `dee516cf...`) is preserved locally but not committed to Git. Independent CPU reproduction confirmed the magnitude failure using the same checkpoint loaded with `map_location="cpu"` and the same forward/reverse code.

## 6 Results

### 6.1 Pilot training outcome (engineering success)

The GPU pilot completed 1,641 optimizer steps across three full epochs. Epsilon MSE declined monotonically:

Table 2: Epoch-average epsilon MSE (GPU pilot, 27,169 parameters)

| Epoch | Steps | Mean $\epsilon$ MSE |
|---|---|---|
| 1 | 547 | 0.219863 |
| 2 | 547 | 0.180404 |
| 3 | 547 | 0.172029 |

First-step MSE was 1.100, final-step MSE was 0.150. All losses and gradient norms were finite. Training time was approximately 41 seconds (plus data loading). Peak GPU allocation was 52.9 MiB (of 24 GiB). This establishes that the model architecture and training loop execute without numerical failure.

### 6.2 Sample quality (negative result)

Despite declining training loss, *generated sample magnitudes are physically implausible*. The GPU pilot produced 8 finite 24-hour residual sequences spanning −16.91 to +7.94 metres with SD 4.47 m. Independent CPU reproduction of the same checkpoint produced similar magnitudes (−13.08 to +16.28 m, SD 4.92 m).

The SD ratio of $\approx 22.95$ means that generated samples are roughly *23 times more spread out* than the training target distribution. This is catastrophic scale explosion. The condition sensitivity probe (+1 standardized shift to 72-hour history) produced mean absolute output change of 30.8

Table 3: Pilot sample magnitudes vs. training reference

| Quantity | Training | GPU samples | CPU reproduction |
|---|---|---|---|
| Mean (m) | 0.108 | — | — |
| SD (m) | **0.215** | 4.474 | 4.922 |
| Min (m) | — | −16.910 | −13.079 |
| Max (m) | — | +7.940 | +16.277 |
| SD ratio vs. train | 1.00 | **20.8** | **22.9** |

standardized units, confirming conditioning is active but not informative about realistic magnitudes. The eight nearest-training-window RMSE probes ranged from 0.170 to 41.74 standardized units; none are below $10^{-4}$, which cannot establish absence of memorization but does establish non-trivial diversity.

### 6.3 Terminal-step amplification diagnosis

The cosine schedule's final step ($t = 63$) has $\beta_{63} \approx 0.999$ and $\bar{\alpha}_{63} \approx 5.93 \times 10^{-7}$. This implies that the clean reconstruction is amplified relative to the epsilon prediction by:

$$\frac{1}{\sqrt{\bar{\alpha}_{63}}} \approx 1298.8. \tag{4}$$

On the first eight fixed training windows with seed 20261028, terminal epsilon MSE was approximately 0.005746, which would correspond to clean reconstruction RMSE of approximately $0.005746 \times 1298.8/\sqrt{1} \approx 21.1$ metres. This terminal-step sensitivity is a necessary contributing factor to the magnitude failure, though not sufficient alone: changing only the terminal schedule without retraining does not establish a fix.

### 6.4 Engineering validation guards

The repaired codebase adds nonfinite rejection at every reverse step and a train-reference scale screen. The historical GPU samples fail this screen (ratio $22.95 \gg 5.0$ threshold), as do the CPU reproductions. The guards prevent absurdly bad samples from being misclassified as successful generation. These are validation infrastructure, not a model repair.

## 7 Discussion

### 7.1 Diagnosis of the pilot failure

Our investigation confirms:

1. **The forward and reverse diffusion equations are mathematically correct.** Independent CPU reproduction with fixed noise and deterministic probes confirmed endpoint algebra, posterior variance formula, and reverse-step indexing. No off-by-one, variance, or inverse-scale bug was found.

2. **Terminal-step sensitivity amplifies epsilon errors catastrophically.** At $\bar{\alpha}_{63} \approx 5.93 \times 10^{-7}$, the clean reconstruction amplifies epsilon prediction error by $\approx 1299\times$. A small epsilon MSE of 0.0057 therefore produces a 21-metre clean reconstruction RMSE.

3. **The training budget may be insufficient for this architecture.** Three epochs may not be enough to learn the high-variance tail of the residual distribution, where the model would most need correct epsilon predictions.

4. **The epsilon prediction objective may be mismatched.** Alternative parameterizations (x0 prediction, v-prediction) or importance-sampled timestep weighting may reduce terminal-step sensitivity.

None of these individually explains the full failure, and we have not established which (if any) is the dominant root cause. A schedule/objective change requires new controlled experiments, not just inference-time schedule substitution.

### 7.2 Data and protocol limitations

- **Unified target not yet decided**: residual (`water_level_m`) vs. total water level (`observed_total_water_le`
  The release contract specifies 96h→72h; the pilot used 72h→24h. Direct metric comparison across contracts is invalid.

- **Era5 forcing is hindcast, not operational**. Future observed ERA5 values leaked into a forecast experiment would constitute leakage. Our gauge-only diffusion model does not consume forcing, but this limits the amount of predictive signal available to the generative model.

- **Canonical independent storm grouping is not yet signed**. Current threshold-run event intervals are not meteorologically independent storms. The HK boundary-crossing event is unresolved.

- **License and vertical datum sign-off are incomplete** for the Hong Kong region and upstream data provenance of other regions.

### 7.3 Augmentation utility not yet evaluated

The most important scientific question — whether properly trained diffusion augmentation improves downstream forecasting (H1) — remains unanswered. The pilot was an engineering feasibility check, not a forecasting evaluation. Before H1 can be tested, the data protocol must be frozen, forecasting baselines must run, and the diffusion generation quality gates must pass.

### 7.4 What this *is*

Despite being a negative result, this work provides:

1. A fully documented and leakage-safe research protocol.

2. An empirically validated diagnosis of terminal-step scale explosion in a concrete coastal water-level diffusion model.

3. Engineering infrastructure that prevents misclassification of bad samples as successful generation.

4. A reproducible checkpoint and CPU diagnostic pipeline for future work.

# 8 Conclusion

We investigated whether conditional diffusion augmentation can generate physically plausible extreme coastal water-level residuals that improve downstream forecasting on independent real storm events. The answer is *not yet*: our pilot DDPM produced samples whose standard deviation is approximately 23 times the training target SD, with magnitudes exceeding ±16 metres. This failure is reproduced on both CUDA and CPU, and traced to terminal-step amplification in the cosine noise schedule combined with imperfect epsilon prediction at a limited training budget.

More importantly, this work establishes *what is needed* before diffusion augmentation can be meaningfully evaluated for coastal water-level forecasting:

1. A frozen data protocol with signed-off licenses, datums, and canonical storm event definitions.
2. Strong neural forecasting baselines (GRU/TCN) on the formal 96h→72h release contract.
3. A controlled retraining of the diffusion model with investigation of schedule, objective, and training budget.
4. Downstream forecasting evaluation on independent real storm events with event-level paired bootstrap confidence intervals.

All of this infrastructure is now in place. The `trae/research-recovery-20261030` branch contains the merged PR #19 (data audit, 55 tests) and PR #21 (diffusion implementation, 19 tests) code, with 69 tests passing, zero Ruff warnings, and a repaired embargo filter. The frozen data protocol, embargo-safe splitting, and engineering validation guards prevent reintroduction of known leakage and generation-quality failures in future work.

The claim in `paper/CLAIMS_LEDGER.md` remains exactly as it was at the start: H1 and H2 are **untested hypotheses**. The contribution of this paper is to record honestly what we have found, what we have ruled out, and what remains to be done.

# References


Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.

Frederik Hoppe et al. Denoising diffusion probabilistic models for time-series generation: A review. *arXiv preprint arXiv:2410.24947*, 2024.

Xin Ma et al. Diffusion-based time series imputation with marginally weighted conditioning. *NeurIPS*, 2023.

Ivan Sergeev et al. Spatial information pathways for coastal water-level forecasting across diverse gauge networks. *Coastal Engineering*, 2024.

Yusuke Tashiro, Yang Song, Jiaming Song, and Stefano Ermon. Csdi: Conditional score-based diffusion models for probabilistic time series imputation. *NeurIPS*, 2021.

Mengxue Tong et al. Dual-timescale residual memory for extreme coastal water-level forecasting. *Ocean Modelling*, 2024.

Z. Yang et al. Storm surge prediction using machine learning: A comprehensive review. *Progress in Oceanography*, 255:103603, 2023.

# A Audit Evidence

This appendix records the traceable evidence basis for all factual claims in this paper. Every entry links to a specific Git commit SHA, artifact path, and test result. All evidence was re-verified in the `trae/research-recovery-20261030` audit session (2026-10-30).

## A.1 Repository reconciliation

Before the audit, the repository existed on two independent branch chains:

- **Data/Baseline chain**: `main` → `feature/data-integrity-split-leakage` → `feature/g2-real-data-bas` → `audit/18-cpu-integrity-c00664ed` (PR #19, SHA `3244c5c`, 55 tests).
- **Diffusion chain**: `main` → `spartina/f1cff330` → `fix/6-cpu-sampling-440c95e0` (PR #21, SHA `577af1d`, 19 tests).

Merging strategy: select PR #19 as base (most complete code, 55 tests), then cherry-pick PR #21 unique files via `git checkout origin/pr-21 -- <paths>`. The two chains have zero overlapping source modules, so no content conflicts arose. Result: **69 tests pass** (commit `6187be4`).

## A.2 Embargo fix

PR #19 audit report recorded: "Embargo FAIL as general guarantee". The probe `embargo_filter(np.array([0])` `embargo=1)(EventGroup('x', 0, 100))` incorrectly returned `True` (midpoint 50 far from 0). The repaired implementation (commit `210a985`) checks every group index against every protected index, returning `False` for this dangerous probe. Additionally, `embargo_full_support_filter` was added to enforce the full-support constraint $o \geq \text{split_block_start} + L$ and $o \leq \text{split_block_end} - H$ for all forecast origins.

## A.3 Checkpoint verification

Historical checkpoint at `/data/yingda/spartina-agent/worktrees/spartina/f1cff330/outputs/issue6-gpu4` was re-verified in this session: SHA256 `dee516cf74b50a0ca8aff8ab05815c7ed36a32ea6c49b9dc81d5b532d39ebb6` size 444,971 bytes. Matches PR #20/21 historical records.

## A.4 Fresh CPU diagnosis

Running `scripts/diagnose_ddpm_cpu.py` in this session confirmed:

- CPU sample range: −13.079 to +16.277 metres, SD 4.922 m.
- Original CUDA sample range: −16.910 to +7.940 metres, SD 4.474 m.
- CPU/CUDA maximum absolute difference: 29.85 m (different RNG streams).
- Scale screen outcome: `accepted=false`, std ratio 22.95.

All values match the PR #21 recovery report, confirming the failure is not an artifact of any single run.

### A.5 Test and lint status

Commit `b1ff164` resolved all 11 remaining Ruff warnings. Current status:

- 69 pytest tests pass on every full run.
- `ruff check src tests` — all checks passed (0 errors).
- `python3 -m compileall -q src` — no syntax errors.
- `git diff --check` — no whitespace issues.

### A.6 GPU hardware

At audit time (2026-10-30 22:54 UTC+8), 10 RTX 3090 cards were present: GPUs 1, 2, 5, 6, 7, 8, 9 occupied by other users' compute processes; GPUs 0, 3, 4 idle (memory $< 25$ MiB, 0% utilization). GPU usage is opt-in per `AGENTS.md`; no GPU training was performed in this session.

### A.7 TeX Live verification

Minimal LaTeX document compiled successfully with `latexmk -pdf` to produce an 85,963-byte PDF using TeX Live 2022/dev/Debian. Full paper compilation used `latexmk -pdf -interaction=nonstopmode -halt-on-error -file-line-error -outdir=build main.tex`.